data_preview

April 29, 2026

1 HW2 Data Preview

```
Fed Funds Futures Raw Table | DataFrame | shape=(2, 9)

                           rtype  publisher_id  instrument_id    open    high  \
ts_event
2026-02-03 00:00:00+00:00     35             1         338078  96.815  96.815
2026-03-20 00:00:00+00:00     35             1         338078  96.665  96.665

                              low   close  volume symbol
ts_event
2026-02-03 00:00:00+00:00  96.815  96.815       2   ZQZ7
2026-03-20 00:00:00+00:00  96.585  96.585       2   ZQZ7

Fed Funds Futures Through Dec 2027 | Series | shape=(2,)

                           Fed Funds Futures Through Dec 2027
ts_event
2026-02-03 00:00:00+00:00                              96.815
2026-03-20 00:00:00+00:00                              96.585

Fed Funds Target Upper Bound | Series | shape=(6344,)

            Fed Funds Target Upper Bound
2008-12-16                          0.25
2008-12-17                          0.25
2008-12-18                          0.25
2008-12-19                          0.25
2008-12-20                          0.25

Fed Funds Target Lower Bound | Series | shape=(6344,)

            Fed Funds Target Lower Bound
2008-12-16                           0.0
2008-12-17                           0.0
2008-12-18                           0.0
2008-12-19                           0.0
2008-12-20                           0.0

S&P 500 ETF Monthly Returns | Series | shape=(8139,)
```

```
            S&P 500 ETF Monthly Returns
date
1993-01-29                    24.358021
1993-02-01                    24.531264
1993-02-02                    24.583210
1993-02-03                    24.843102
1993-02-04                    24.947049
```

High Yield Bond ETF Returns | Series | shape=(4564,)

```
            High Yield Bond ETF Returns
date
2007-04-11                    35.563918
2007-04-12                    35.587775
2007-04-13                    35.523020
2007-04-16                    35.509388
2007-04-17                    35.492347
```

Commodity ETF Returns | Series | shape=(4859,)

```
            Commodity ETF Returns
date
2006-02-06              21.339430
2006-02-07              20.722174
2006-02-08              20.633994
2006-02-09              20.827989
2006-02-10              20.448729
```

10-Year Treasury Yield | Series | shape=(9086,)

```
            10-Year Treasury Yield
1990-01-02                  0.0794
1990-01-03                  0.0799
1990-01-04                  0.0798
1990-01-05                  0.0799
1990-01-08                  0.0802
```

US Bond Monthly Returns | Series | shape=(435,)

```
            US Bond Monthly Returns
1990-02-28                 0.001691
1990-03-31                -0.002210
1990-04-30                -0.018546
1990-05-31                 0.036101
1990-06-30                 0.018418
```

Combined Asset Return Table | DataFrame | shape=(435, 4)

```
           US Equity   US Bond Commodities US HY Credit
1990-02-01            0.001691
1990-03-01           -0.002210
1990-04-01           -0.018546
```

```
1990-05-01            0.036101
1990-06-01            0.018418
```

Fed Funds Midpoint Table | Series | shape=(861,)

```
            Fed Funds Midpoint Table
1954-07-01                      0.80
1954-08-01                      1.22
1954-09-01                      1.07
1954-10-01                      0.85
1954-11-01                      0.83
```

Data Availability Cut Mask | Series | shape=(861,)

```
            Data Availability Cut Mask
1954-07-01                       False
1954-08-01                       False
1954-09-01                       False
1954-10-01                       False
1954-11-01                       False
```

Joined Macro and Asset Table | DataFrame | shape=(434, 5)

```
           US Equity   US Bond Commodities US HY Credit  cutting
1990-02-01            0.001691                              True
1990-03-01           -0.002210                              True
1990-04-01           -0.018546                              True
1990-05-01            0.036101                             False
1990-06-01            0.018418                             False
```

Sample After Availability Cut | DataFrame | shape=(95, 4)

```
           US Equity   US Bond Commodities US HY Credit
1990-02-01            0.001691
1990-03-01           -0.002210
1990-04-01           -0.018546
1990-12-01            0.018975
1991-01-01            0.010119
```

Sample Before Availability Cut | DataFrame | shape=(339, 4)

```
           US Equity   US Bond Commodities US HY Credit
1990-05-01            0.036101
1990-06-01            0.018418
1990-07-01            0.011692
1990-08-01           -0.026474
1990-09-01            0.010001
```

Macro Regime Summary Table | DataFrame | shape=(4, 6)

```
              Cutting Ann. Ret %  Cutting Ann. Vol %  Cutting % months > 0  \
US Equity                  -6.62               20.19                  31.6
US Bond                     6.37                8.40                  62.1
Commodities               -14.72               26.40                  17.9
```

```
US HY Credit                -3.31                18.99                   14.7

              Non-Cutting Ann. Ret %  Non-Cutting Ann. Vol %  \
US Equity                      15.46                   13.36
US Bond                         4.65                    7.10
Commodities                     5.36                   16.44
US HY Credit                    6.85                    7.75

              Non-Cutting % months > 0
US Equity                         65.2
US Bond                           57.2
Commodities                       28.9
US HY Credit                      36.3

Forward Return Table | DataFrame | shape=(7, 4)

        US Equity   US Bond Commodities US HY Credit
1990-07            0.098024
1995-07  0.259865  0.020953
1998-09  0.392659 -0.023805
2001-01 -0.117523  0.050488
2007-09 -0.109878  0.098412    0.514812    -0.012181

CPI Level Index | Series | shape=(950,)

            CPI Level Index
1947-01-01            21.48
1947-02-01            21.62
1947-03-01            22.00
1947-04-01            22.00
1947-05-01            21.95

Industrial Production Index | Series | shape=(1287,)

            Industrial Production Index
1919-01-01                       4.8739
1919-02-01                       4.6585
1919-03-01                       4.5238
1919-04-01                       4.6046
1919-05-01                       4.6315

CPI Year-over-Year Growth | Series | shape=(938,)

            CPI Year-over-Year Growth
1948-01-01                   0.102421
1948-02-01                   0.094820
1948-03-01                   0.068182
1948-04-01                   0.082727
1948-05-01                   0.093850

Industrial Production Year-over-Year Growth | Series | shape=(1275,)
```

```
            Industrial Production Year-over-Year Growth
1920-01-01                                     0.154702
1920-02-01                                     0.208093
1920-03-01                                     0.220235
1920-04-01                                     0.134496
1920-05-01                                     0.156990
```

Macro Regime Labels | DataFrame | shape=(374, 5)

```
             CPI YoY    IP YoY  HighGrowth  HighInflation         regime
1995-01-01  0.028708  0.068005        True           True  GrowUp/InflUp
1995-02-01  0.028630  0.066178        True           True  GrowUp/InflUp
1995-03-01  0.027872  0.057165        True           True  GrowUp/InflUp
1995-04-01  0.031250  0.049136        True           True  GrowUp/InflUp
1995-05-01  0.031186  0.049460        True           True  GrowUp/InflUp
```

Regime Spell Table | DataFrame | shape=(60, 4)

```
          regime      start        end  months
0  GrowUp/InflUp 1995-01-01 1995-12-01      12
1  GrowDn/InflUp 1996-01-01 1996-01-01       1
2  GrowUp/InflUp 1996-02-01 1997-03-01      14
3  GrowUp/InflDn 1997-04-01 1999-08-01      29
4  GrowUp/InflUp 1999-09-01 2000-10-01      14
```

Combined Analysis Table | DataFrame | shape=(374, 5)

```
            US Equity   US Bond Commodities US HY Credit         regime
1995-01-01   0.033607  0.022958                           GrowUp/InflUp
1995-02-01   0.040810  0.032618                           GrowUp/InflUp
1995-03-01   0.027841  0.007424                           GrowUp/InflUp
1995-04-01   0.029623  0.015155                           GrowUp/InflUp
1995-05-01   0.039673  0.060437                           GrowUp/InflUp
```

Performance Summary Table | DataFrame | shape=(4, 12)

```
               Ann Mean %                                      Ann Vol %  \
Asset         Commodities   US Bond  US Equity US HY Credit Commodities
regime
GrowDn/InflDn    1.818714  3.476372  13.830481    10.411836   19.091060
GrowDn/InflUp   -6.382620  1.402353   0.068144    -0.967345   19.460016
GrowUp/InflDn   -2.859281  5.909888  16.180316     5.313238   16.626181
GrowUp/InflUp   18.599262  6.509994  12.439136    -0.167377   17.775279

                                                   Sharpe            \
Asset           US Bond  US Equity US HY Credit Commodities   US Bond
regime
GrowDn/InflDn  8.340451  15.911518    12.179009    0.095265  0.416809
GrowDn/InflUp  7.625240  17.142205    11.669200   -0.327986  0.183909
GrowUp/InflDn  6.874179  14.988998     7.048034   -0.171975  0.859723
GrowUp/InflUp  6.934548  12.636820     8.534279    1.046356  0.938777
```

```
Asset         US Equity US HY Credit
regime
GrowDn/InflDn  0.869212     0.854900
GrowDn/InflUp  0.003975    -0.082897
GrowUp/InflDn  1.079479     0.753861
GrowUp/InflUp  0.984356    -0.019612
```

```
Daily S&P 500 Returns | Series | shape=(6412,)

            Daily S&P 500 Returns
date
1999-12-01              88.653199
1999-12-02              89.185976
1999-12-03              90.823654
1999-12-06              90.152786
1999-12-07              89.422753
```

```
Weekly S&P 500 Returns | Series | shape=(1330,)

            Weekly S&P 500 Returns
date
1999-12-10               -0.013686
1999-12-17                0.008180
1999-12-24                0.026609
1999-12-31                0.002667
2000-01-07               -0.007660
```

```
Daily Treasury Yield Changes | Series | shape=(6605,)

            Daily Treasury Yield Changes
1999-12-01                        0.0621
1999-12-02                        0.0624
1999-12-03                        0.0617
1999-12-06                        0.0616
1999-12-07                        0.0611
```

```
Weekly Treasury Yield Changes | Series | shape=(1379,)

            Weekly Treasury Yield Changes
1999-12-03                         0.0617
1999-12-10                         0.0608
1999-12-17                         0.0630
1999-12-24                         0.0641
1999-12-31                         0.0645
```

```
Weekly Bond Returns | Series | shape=(1378,)

            Weekly Bond Returns
1999-12-10             0.007829
1999-12-17            -0.015136
1999-12-24            -0.006859
```

```
1999-12-31            -0.001687
2000-01-07            -0.003860
```

Weekly Stock and Bond Return Table | DataFrame | shape=(1330, 2)

```
              Equity      Bond
1999-12-10 -0.013686  0.007829
1999-12-17  0.008180 -0.015136
1999-12-24  0.026609 -0.006859
1999-12-31  0.002667 -0.001687
2000-01-07 -0.007660 -0.003860
```

2000 to 2019 Weekly Sample | DataFrame | shape=(1043, 2)

```
              Equity      Bond
2000-01-07 -0.007660 -0.003860
2000-01-14  0.008362 -0.011093
2000-01-21 -0.017223 -0.005920
2000-01-28 -0.059282  0.010632
2000-02-04  0.049448  0.010663
```

Rolling Volatility Inputs | Series | shape=(1330,)

```
           Rolling Volatility Inputs
1999-12-10
1999-12-17
1999-12-24
1999-12-31
2000-01-07
```

2000 to 2019 Rolling Volatility Inputs | Series | shape=(996,)

```
            2000 to 2019 Rolling Volatility Inputs
2000-12-01                                0.009982
2000-12-08                                0.027161
2000-12-15                                0.025558
2000-12-22                                0.046820
2000-12-29                                0.045350
```

Equity Rolling Volatility | Series | shape=(1330,)

```
           Equity Rolling Volatility
1999-12-10
1999-12-17
1999-12-24
1999-12-31
2000-01-07
```

Bond Rolling Volatility | Series | shape=(1330,)

```
           Bond Rolling Volatility
1999-12-10
1999-12-17
1999-12-24
```

```
1999-12-31
2000-01-07

Equity Portfolio Weights | Series | shape=(1330,)

           Equity Portfolio Weights
1999-12-10
1999-12-17
1999-12-24
1999-12-31
2000-01-07

Bond Portfolio Weights | Series | shape=(1330,)

           Bond Portfolio Weights
1999-12-10
1999-12-17
1999-12-24
1999-12-31
2000-01-07

Risk Parity Portfolio Returns | Series | shape=(1278,)

            Risk Parity Portfolio Returns
2000-12-08                       0.013822
2000-12-15                       0.005293
2000-12-22                       0.011537
2000-12-29                      -0.004861
2001-01-05                       0.008707

2000 to 2019 Equity Weights | Series | shape=(995,)

            2000 to 2019 Equity Weights
2000-12-08                     0.224867
2000-12-15                     0.227179
2000-12-22                     0.222082
2000-12-29                     0.224562
2001-01-05                     0.226252

2000 to 2019 Bond Weights | Series | shape=(995,)

            2000 to 2019 Bond Weights
2000-12-08                   0.775133
2000-12-15                   0.772821
2000-12-22                   0.777918
2000-12-29                   0.775438
2001-01-05                   0.773748

2000 to 2019 Portfolio Returns | Series | shape=(995,)

            2000 to 2019 Portfolio Returns
2000-12-08                        0.013822
2000-12-15                        0.005293
2000-12-22                        0.011537
```

```
2000-12-29                       -0.004861
2001-01-05                        0.008707
```

```
Equity Risk Contribution | Series | shape=(1043,)
```

```
           Equity Risk Contribution
2000-01-07
2000-01-14
2000-01-21
2000-01-28
2000-02-04
```

```
Bond Risk Contribution | Series | shape=(1043,)
```

```
           Bond Risk Contribution
2000-01-07
2000-01-14
2000-01-21
2000-01-28
2000-02-04
```

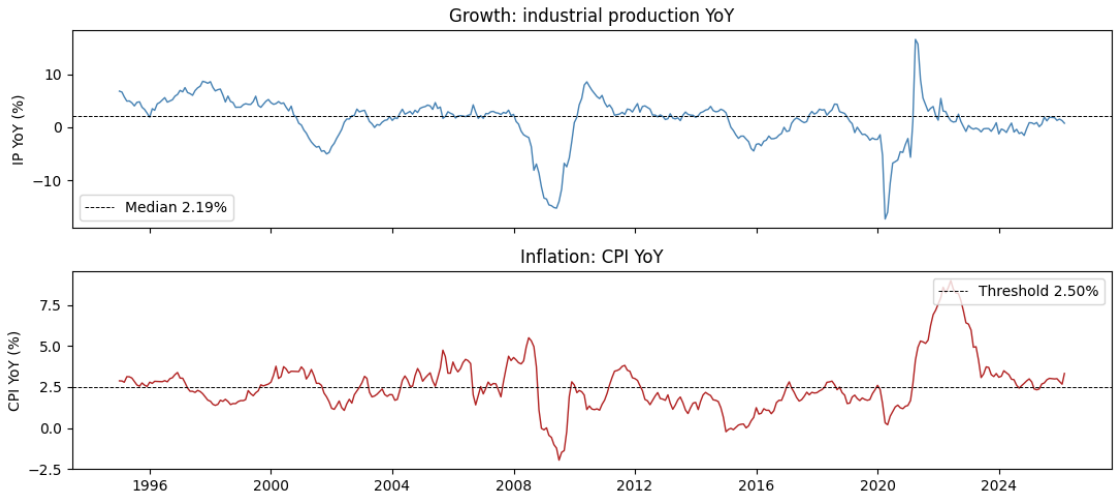
```
Rolling Portfolio Volatility | Series | shape=(1278,)
```

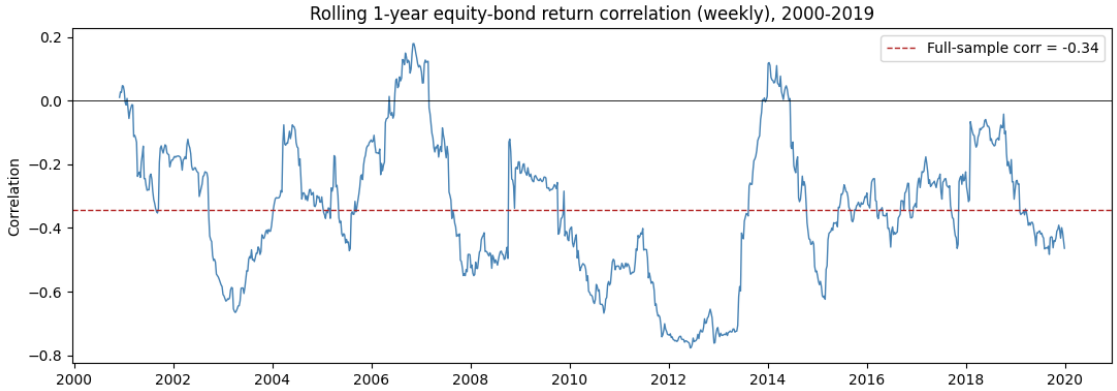
```
           Rolling Portfolio Volatility
2000-12-08
2000-12-15
2000-12-22
2000-12-29
2001-01-05
```

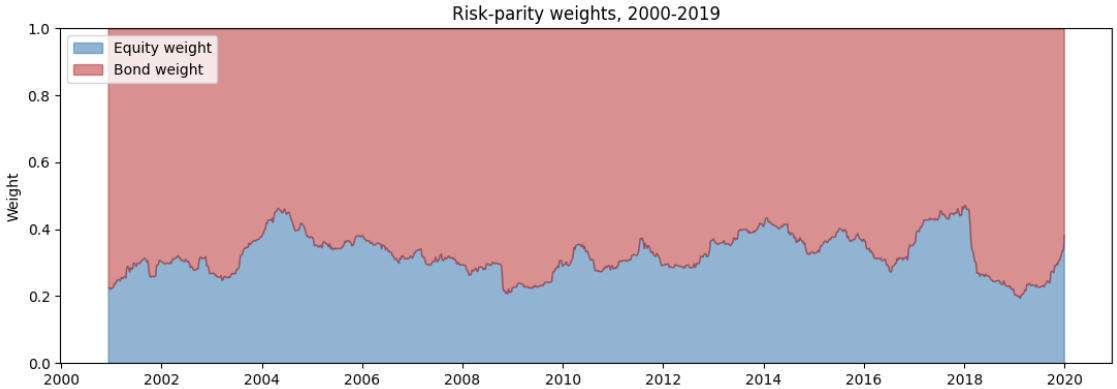
```
Full-Sample Rolling Volatility Table | Series | shape=(1279,)
```

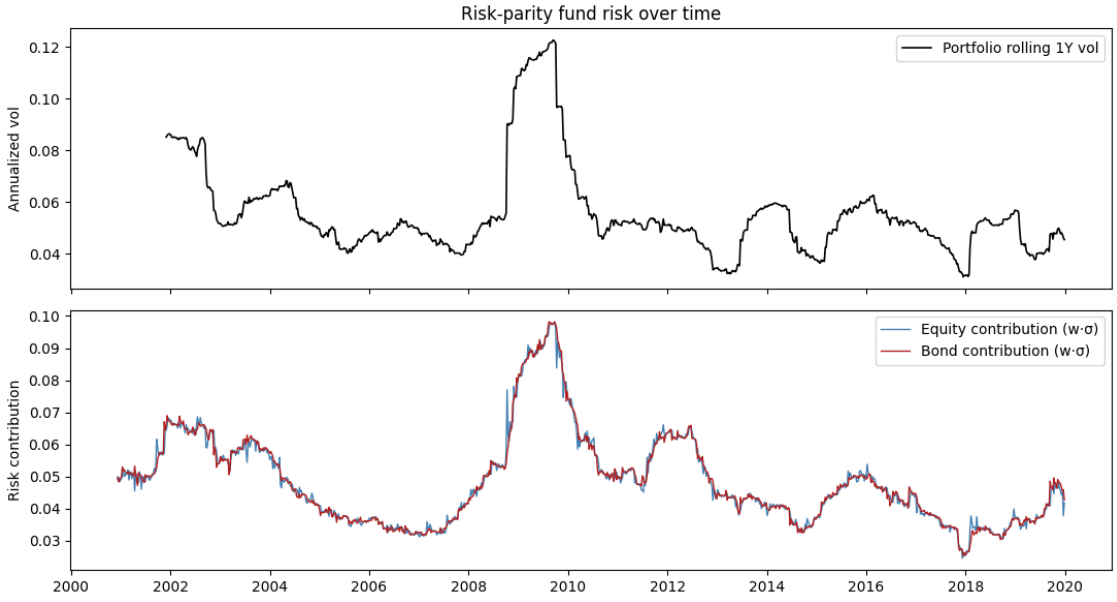
```
            Full-Sample Rolling Volatility Table
2000-12-01                              0.009982
2000-12-08                              0.027161
2000-12-15                              0.025558
2000-12-22                              0.046820
2000-12-29                              0.045350
```

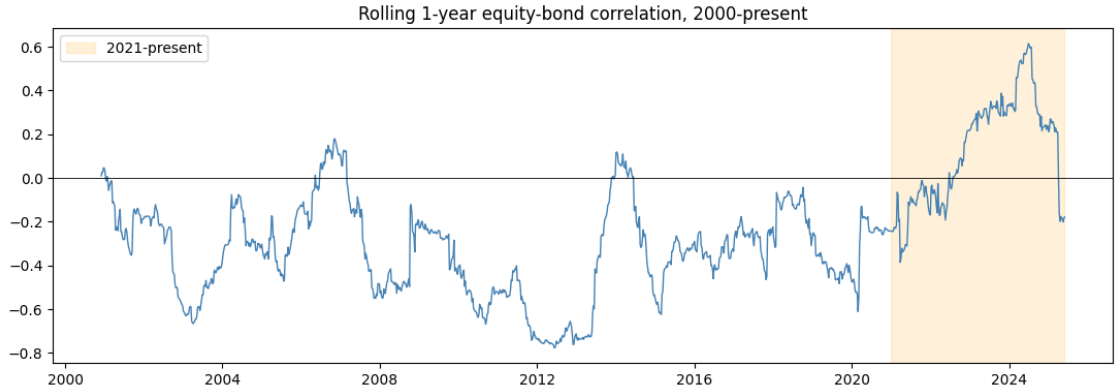
1.1 Graphs


Growth: industrial production YoY
IP YoY (%)
Median 2.19%
Inflation: CPI YoY
CPI YoY (%)
Threshold 2.50%


Rolling 1-year equity-bond return correlation (weekly), 2000-2019
Correlation
Full-sample corr = -0.34


Risk-parity weights, 2000-2019
Weight
Equity weight
Bond weight


Risk-parity fund risk over time
Portfolio rolling 1Y vol
Annualized vol
0.12
0.10
0.08
0.06
0.04
Equity contribution (w·σ)
Bond contribution (w·σ)
Risk contribution
0.10
0.09
0.08
0.07
0.06
0.05
0.04
0.03
2000
2002
2004
2006
2008
2010
2012
2014
2016
2018
2020


Rolling 1-year equity-bond correlation, 2000-present
2021-present
0.6
0.4
0.2
0.0
−0.2
−0.4
−0.6
−0.8
2000
2004
2008
2012
2016
2020
2024